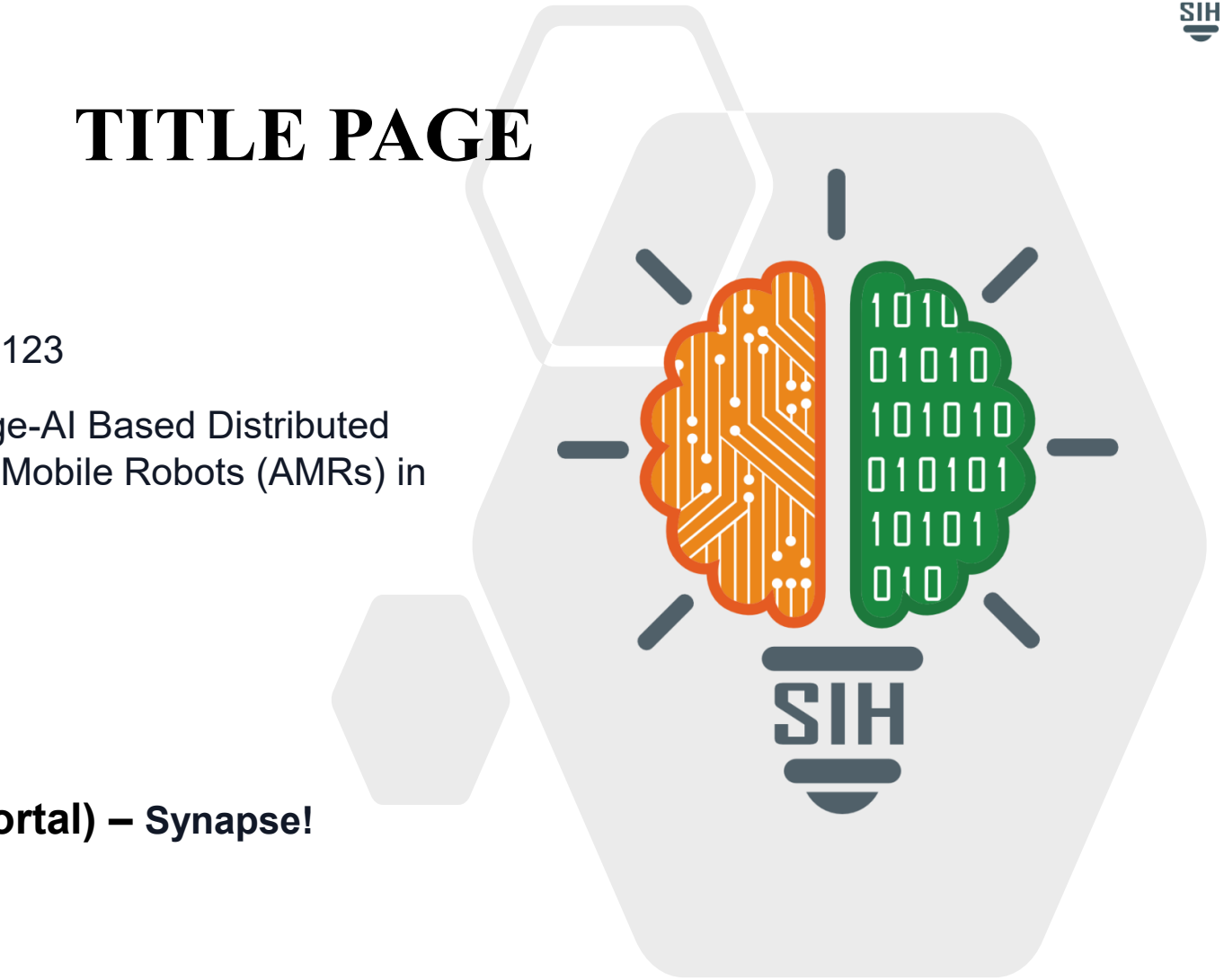
TITLE PAGE

- **Problem Statement ID –** SIH26123
- **Problem Statement Title –** Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID – 168339**
- **Team Name (Registered on portal) – Synapse!**



Synapse!

SANCHAR: an AMR fleet with no master, that keeps moving when the network fails

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Proposed Solution

Every AMR runs the identical edge node. It plans its own path, negotiates right-of-way with its peers over a signed peer-to-peer radio protocol, and bids for orders in a leaderless market. There is no server, no leader and no single point of failure.

Detailed explanation of the proposed solution

- Peer-to-peer mesh: robots broadcast position, battery, intended path and next-node claim twice every second, relayed over multiple hops.
- Guided-PIBT motion coordination, the algorithm family that won the 2023 League of Robot Runners, made fully decentralized with swap resolution and corridor gates for choke points.
- Leaderless order market: auction bids with consensus-free awards, so orders are re-assigned when an aisle is blocked or a robot dies.
- Fleet dashboard that only listens: live positions, battery, plans and the reason for every move.
- Decentralized network stack for inter-robot messaging: peers share localization (pose) and intent.

How it addresses the problem

- No central server, so no latency to a server and no single point of failure.
- Claim-before-move safety rule: zero inter-robot collisions in all 330 benchmark runs.
- 25% faster task completion than stop-and-wait on overlapping paths (PS target: 20%).
- Wi-Fi dead zones: short-range robot-to-robot links keep the fleet working. A robot cut off completely stops safely.
- Real-time conflict resolution and collision avoidance: multi-agent path finding (MAPF) with a wait-for cycle breaker keeps choke points deadlock-free.
- Task allocation (MRTA) and re-routing: pickup points are re-assigned by auction, and robots re-route around blocked aisles.

Innovation and uniqueness of the solution

- Acknowledged-motion rule: collision-freedom holds under packet loss, relays and robot crashes.
- Partition-tolerant task board (CRDT): network islands re-merge with exactly one owner per order.
- Explainable: every push, swap, yield and re-assignment is logged with its reason.
- Measured honestly: 30 seeds per scenario, 95% confidence intervals, and our weak case reported.

0
inter-robot collisions in 330 runs

25%
faster than stop-and-wait, 6 AMRs (PS target 20%)

43%
faster with live orders and self-allocation

0
servers: no single point of failure

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TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

- Python 3.12 edge node with no heavy dependencies. Runs on Raspberry Pi 4/5 or Jetson (ARM64).
- msgspec binary messages, HMAC-SHA256 signing (Ed25519 in production), anti-replay window.
- Radio: UDP multicast over 802.11 ad-hoc or ESP-NOW; optional ROS 2 bridge via rmw_zenoh.
- Planning: windowed space-time A*, decentralized PIBT with swap resolution, corridor gates.
- Order market: sequential single-item auction with a CRDT task board.
- Dashboard: standard-library web server with live Server-Sent Events and an HTML5 Canvas map.
- Edge-AI on every robot: AI planning and negotiation (A*, PIBT, auctions) plus a learned travel-time predictor from radio-derived congestion features: 10% lower ETA error; bidding with it is 4.6% faster on a 12-robot floor (14/20 seeds).
- Verification: pytest and Hypothesis property tests; 30-seed benchmark harness against stop-and-wait.
- Motion: discrete-time kinematic model on a roadmap graph (1 m cells, 1 m/s, one node per 1 s tick). Next: continuous velocity and acceleration profiles for differential-drive robots with real footprints, under the same safety rule.

Methodology and process for implementation (Flow Charts/Images/ working prototype)

1. Sense and gossip

Pose, battery, sightings and orders; signed broadcasts, relayed by peers

2. Plan guide path

Space-time A* around higher-priority robots' broadcast plans

3. Claim next node

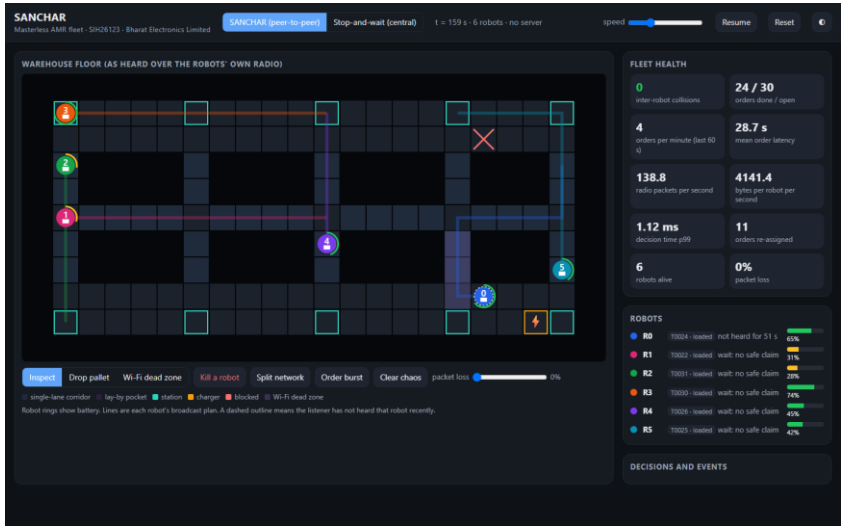
PIBT push, swap, yield or corridor gate decide the claim

4. Safety check

Claim heard by all? Peers' acks recent? Any conflicting claim?

5. Move or wait, then report

Order market: bid, claim, release or re-open; dashboard listens



Working prototype (live dashboard)

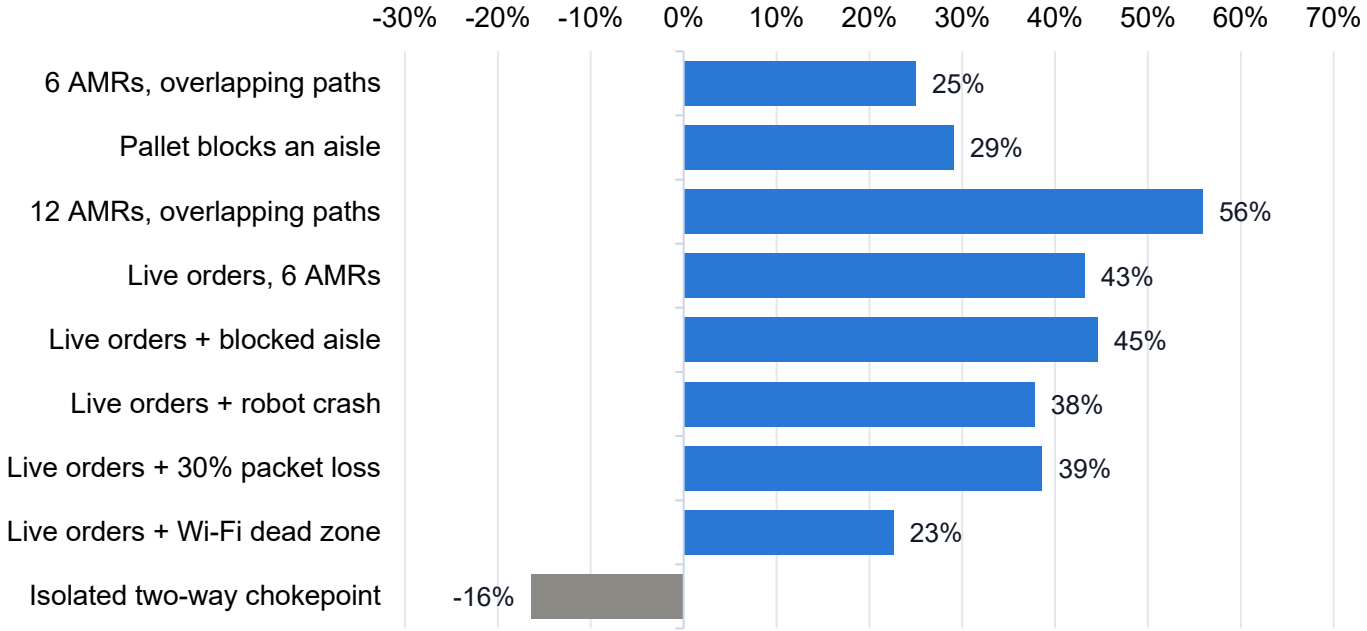
- The map shows what a passive radio listener hears: robots, battery rings, broadcast plans, a dropped pallet and a Wi-Fi dead zone.
- Live chaos controls: drop a pallet, paint a dead zone, kill a robot, split the network, set packet loss.
- Toggle to stop-and-wait to compare on the same floor and orders.
- Multi-process mode: one OS process per robot over real UDP. With robot R2's process killed mid-run, the other 5 delivered all 24/24 orders with 0 collisions.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea: makespan reduction vs stop-and-wait (30 seeds each)



- Zero collisions and zero unfinished runs for SANCHAR in all 330 runs. Stop-and-wait gridlocked in 70; at a single-lane 4-way intersection it gridlocked in 30/30 runs.
- Edge-ready: each decision takes under 2 ms (p99) on a laptop core, with about 2.5 KB of radio traffic per robot per second.
- Bars show seeds where both methods finished; the grey bar is our weak case, reported honestly.
- The prototype is a multi-robot simulation running the edge code planned for Raspberry Pi 4/5 and Jetson Nano; on hardware, pose comes from each robot's own localization (SLAM or odometry), and its LiDAR safety scanner remains the physical backstop.

Potential challenges and risks → Strategies for overcoming these challenges

Packet loss, radio interference

Acknowledged motion plus multi-hop relays: still zero collisions at 30–50% loss; 39% faster than stop-and-wait even at 30% loss

Wi-Fi dead zones in metal racking

Direct robot-to-robot links keep robots talking. A fully cut-off robot stops within 2 s and resumes on contact

Robot breakdown mid-task

Its orders are re-opened and re-auctioned automatically. Its body is sensed and routed around

Blocked aisle or station

Route around it; release and re-auction orders; redirect deliveries to the nearest free station

Isolated two-way chokepoint

Here we are 16% slower than ideal central zone control. Planned: direction batching at choke points

Scale to 100+ robots

Messages grow linearly with fleet size and planning is local; 12-robot runs are 56% faster than stop-and-wait

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IMPACT AND BENEFITS

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Potential impact on the target audience

- BEL and defence logistics: ordnance depots, naval dockyards and forward supply bases, where there may be no network infrastructure, radio silence or jamming, and a fleet must keep working without a server.
- BEL's own factories and stores: intralogistics that tolerates metal-racking dead zones.
- Indian warehouse operators and robot makers (such as Addverb and GreyOrange), 3PLs and MSMEs: fleet software without a central server to license, host and keep alive.
- Floor staff: robots that explain their decisions and never collide, a safer shared workplace.
- Operations teams: throughput holds up when things go wrong; a blocked aisle, dead zone or robot failure slows the fleet instead of stopping it.

Benefits of the solution (social, economic, environmental, etc.)

- Economic: 43% shorter completion time and 62% lower order latency than stop-and-wait with a central dispatcher, from the same robots.
- Economic: no fleet server or cloud subscription. Unplanned warehouse downtime is often costed at about $125K per hour, and there is no longer a single point of failure to cause it.
- Operational: zero collisions; the fleet keeps delivering through blocked aisles, dead zones and robot failures.
- Strategic and social: an indigenous, Make-in-India coordination stack for critical sectors, with no foreign cloud dependency.
- Environmental: smoother flow means less stop-and-go and idle time per order, and battery-aware bidding sends robots to charge before they run flat.
- Aligned with SDG 9 (industry, innovation, infrastructure) and SDG 8 (decent work).

43%
shorter completion time with live orders

62%
lower order latency

38%
faster even when a robot breaks down

0
collisions, servers or cloud dependency

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

Research

- Okumura et al., Priority Inheritance with Backtracking (PIBT), Artificial Intelligence, 2022
- Chen, Harabor, Li, Stuckey, Traffic Flow Optimisation for Lifelong MAPF, AAAI 2024
- Jiang et al., Scaling Lifelong MAPF (WPPL), winner of the League of Robot Runners 2023
- Okumura, LaCAM: Search-Based Algorithm for Quick MAPF, AAAI 2023
- Li et al., Lifelong MAPF in Large-Scale Warehouses (RHCR), AAAI 2021
- Li et al., Graph Neural Networks for Decentralized Multi-Robot Path Planning, IROS 2020
- Choi, Brunet, How, Consensus-Based Decentralized Auctions for Task Assignment (CBBA), IEEE T-RO 2009
- Shapiro et al., Conflict-free Replicated Data Types, SSS 2011
- Hönig et al., Multi-Agent Path Finding with Kinematic Constraints, ICAPS 2016

Standards, industry and prior work

- VDA 5050 AGV/AMR interface standard; MassRobotics AMR Interoperability Standard
- ROS 2 rmw_zenoh: github.com/ros2/rmw_zenoh
- Addverb Movect fleet manager; EE Times, Addverb brings AI and edge computing to warehouse robotics
- BEL and DRDO R&DE(E): Unexploded Ordnance Handling Robot licence (bel-india.in)
- Competitor prototypes reviewed for SIH26123 (GitHub): pure-Python grid sims and ROS 2/Gazebo stacks. None combines provable safety, a leaderless order market, crash and dead-zone tolerance and a 30-seed benchmark
- Our prototype, benchmark and tests: sanchar repository, https://github.com/heetdaryani7907/sanchar-sih26123

Scan for the code, benchmark and tests

How SANCHAR compares (✓ yes · ◐ partly · ✗ no)

Capability	Central fleet manager (industry today)	Public SIH26123 prototypes we reviewed	SANCHAR
Keeps working if the server or Wi-Fi fails	✗	◐	✓
Collision-freedom argued and tested under packet loss	◐	✗	✓
Orders re-assigned without a central dispatcher	✗	◐	✓
Survives robot crash, blocked aisle and dead zone	◐	◐	✓
Benchmarked on 30 seeds per scenario with confidence intervals	–	✗	✓